SiamBAG: Band Attention Grouping-Based Siamese Object Tracking Network for Hyperspectral Videos

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***Abstract*— A hyperspectral video contains frames with numerous spectral bands, providing fine reflectance information for object identification and tracking. Enriched features can be learned from spectral–spatial data using deep learning models. However, due to the difficulty in hyperspectral video collection, deep model training is often insufficient, causing reduced performance during the testing stage. To address this issue, we present a novel band attention grouping-based Siamese framework (SiamBAG) for hyperspectral object tracking. SiamBAG uses massive color object tracking data to train a deep neural network. Band weights obtained by the band attention module are used to group a hyperspectral image (HSI) into multiple three-channel false-color images with approximate total group weights. Then multiple enhanced images obtained by histogram equalization are fed to the proposed SiamBAG network to generate a classification branch, a regression branch, and a scale-tuning branch. In the classification branch, the response maps of multiple groups are fused by regularized group weights to estimate the position of objects. Then the regression branch is used to obtain the initial object position of objects. The position offsets are fed back to the scale tune branch to relocate and fine-tune the object position by exploiting the similarity between template features and detection features. Experimental results demonstrate that the proposed tracker achieves superior tracking performance than other methods. The source codes of this article will be released at https://github.com/zephyrhours/Hyperspectral-Object-Tracking-SiamBAG.**

***Index Terms*— Correlation filter, hyperspectral video, object tracking, scale adaptive, tensor.**

I. Introduction

As a fundamental task of computer vision, object tracking aims to predict the motion state of an object in the video sequences [1], [2]. It has been widely used in applications such as human–computer interaction, virtual reality, intelligent video surveillance, robot visual navigation, autonomous driving, and intelligent traffic control [3], [4], [5]. Different from color object trackers that rely on red, green and blue (RGB) videos for object tracking, hyperspectral trackers use videos containing hyperspectral images (HSIs) with numerous spectral bands. These HSIs are favorable to object tracking by providing fine spectral "diagnostic" information because each material has unique spectral features [6].

Hyperspectral object trackers can be roughly classified into *generative models* and *discriminative models*. Early hyperspectral trackers were mainly generative models, which used early video sequence data to generate a statistical model of the tracking object, and then performed object discrimination in the subsequent sequence. Yuhas et al. [7] exploited spectral angle mapper (SAM) to distinguish foreground and background [8]. Considering radiation theory and particle filter, Van Nguyen et al. proposed a framework that combines spectral reflectance and a mean-shift tracker to track the object in pseudo-hyperspectral video [9]. However, these generative models only focused on the object itself, while the surrounding background information is ignored, thus producing inferior tracking performance in challenging scenarios. More recent hyperspectral trackers are mainly based on discriminative models, which can be further divided into two categories: correlation filtering (CF)-based methods and deep-learning-based methods.

A. CF-Based Object Tracking

Qian et al. [10] proposed a convolutional network-based hyperspectral tracking (CNHT) method by extracting 3-D patches from the object region as fixed convolution filters for encoding local spectral–spatial information. To excavate material spatial–spectral information, a local spectral–spatial histogram of multidimensional gradients (SSHMGs) and spatial distribution of materials method was proposed (abbreviated as MHT) [2] to describe the 3-D local spectral–spatial structure in hyperspectral video data. Leveraging the advantage of the CF-based tracker that does not require training [11], [12] in object tracking, a CF-based fast spatial–spectral features (FSSFs) [13] method was presented to implement robust and real-time object tracking in hyperspectral surveillance video, where a spatial–spectral convolution kernel was used to encode the discriminative information. To effectively extract the spatial–spectral feature directly from the mosaic spectral images, a feature descriptor using histogram of oriented mosaic gradient (HOMG) was developed, where two forms of spectral filter arrays neighborhood (SFAN) were found to construct a feature extractor suitable for any spectral filter

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arrays' structure [14]. Tang et al. [15] constructed a joint hyperspectral tracking framework (TASSCF) to exploit spectral and spatial information by integrating the target-aware band selection method and adaptive spatial–spectral discriminant analysis method. Hou et al. [16] designed a spatial–spectral weighted and regularized the tensor sparse correlation filter (TSCFW) method for hyperspectral object tracking, where tensor represention and 2-D adaptive scale search strategy were first introduced into hyperspectral object tracking.

B. Deep-Learning-Based Object Tracking

Considering the excellent performance of deep features in traditional object tracking, some deep learning methods were proposed for object tracking on hyperspectral video data. Uzkent et al. [17] developed a hyperspectral likelihood map-aided tracking method (HLT) to build offline classifiers instead of learning a generative object model in an online manner for hyperspectral channels. Subsequently, Uzkent et al. [18] proposed a deep hyperspectral kernel correlation filter-based tracker (DeepHKCF) to efficiently track aerial vehicles, where kernelized correlation filters (KCFs) and deep convolutional neural network features were applied. Liu et al. [1] developed an anchor-free hyperspectral target tracking network (HANet), where the entire network was trained end-to-end, and high-level semantic features were exploited for target tracking. Subsequently, a high spectral–spatial temporal resolution hyperspectral video object tracking framework (H^3Net) [19] was proposed, where an end-to-end spectral classification branch was introduced to the anchor-free Siamese network, and the model was trained by the "cycle consistency" principle to increase the network's discriminative ability.

These deep-learning-based methods achieved superior tracking performance by linear and nonlinear transformation (that is, average pooling operation and activation operation) to obtain advanced features of objects. However, the performance of these trackers is limited due to a small number of training samples. To tackle this problem, Li et al. [20] developed a band-attention-aware ensemble network (BAENet) to split an HSI into multiple three-channel images according to their bandwise importance determined by a band attention model. These false-color images were then tracked by a set of weak visual tracking via adversarial learning (VITAL) trackers [21], respectively, which were combined via ensemble learning to determine the location of the object. However, each weak tracker of BAENet only tracks partial object spectral information, making the extracted features not highly discriminative. A material-guided Siamese fusion network (SiamF) [22] was developed to fill this gap by integrating a hyperspectral feature fusion module, an online spatial classifier module, and an online material classifier module to fuse multiple false-color image features. Subsequently, Tang et al. [23] presented a deep hyperspectral tracker based on Siamese network (SiamHT), where the heterogeneous encoder–decoder module and spectral semantic representation module were designed to extract the spatial and spectral semantic features, respectively.

The deep-learning-based trackers are essentially driven by big data, and whether the training dataset is rich or not influences their tracking performance. Models trained with limited data are often overfit, which leads to poor generalization ability. Moreover, existing models were usually only trained and tested on hyperspectral videos, while rich resources of color video datasets were not leveraged to help the training.

Addressing this issue, Liu et al. [24] proposed a dual deep Siamese network framework (SiamHYPER) to learn a hyperspectral tracker from a pretrained RGB tracker. Similarly, Zhao et al. [25] proposed a transformer-based fusion tracking network (TFTN) to fuse hyperspectral and RGB data, where a self-attention module of Transformer was used to combine the information of different modalities for multimodality fusion in tracking. Subsequently, Zhao et al. [26] developed a reliability-guided aggregation network (RANet) for hyperspectral and RGB tracking, which guided the combination of hyperspectral information and RGB information through modality reliability to improve tracking performance.

To explore better semantic features in limited training data, Lei et al. [27] designed a spatial–spectral cross correlation embedded dual-transfer network (SSDTNet) to mine semantic information with limited annotation samples, where transfer learning was used to transfer the knowledge from traditional color videos for hyperspectral tracking. Subsequently, Tang et al. [28] designed a background-aware hyperspectral tracking (BAHT) method, where a background-aware band selection module was used to preserve bands, and then the selected bands were input into the backbone networks pretrained on color videos to describe the deep semantic features of the object. However, for these mentioned methods, these RGB videos are acquired in a close viewpoint as hyperspectral videos, which have the same scene and tracking objects. When the RGB data of the same scene cannot be obtained, it is difficult to guarantee the tracking performance of these models. In addition, hyperspectral training data are still involved in the training process of the model. Thereby, without hyperspectral training data, these methods still cannot perform the tracking task.

To fill in this gap of data scarcity, a band attention grouping-based Siamese object tracking (SiamBAG) framework is proposed for the hyperspectral tracking task. Without hyperspectral training data, the tracking model is trained using a large amount of color data, which cleverly solves the issue of data dependency for hyperspectral tracking tasks.

The main contributions of this article are summarized as follows.

1) A novel band attention grouping-based Siamese hyperspectral object tracking framework is proposed, where model training is completed using massive color object tracking data without the participation of hyperspectral training data, thereby effectively alleviating the impact of data scarcity on training.
2) Relying on the band weights obtained by a band attention module, different hyperspectral bands are redivided into multiple three-channel data with an optimized grouping strategy, which ensures the approximate group weights of each three-channel data to balance

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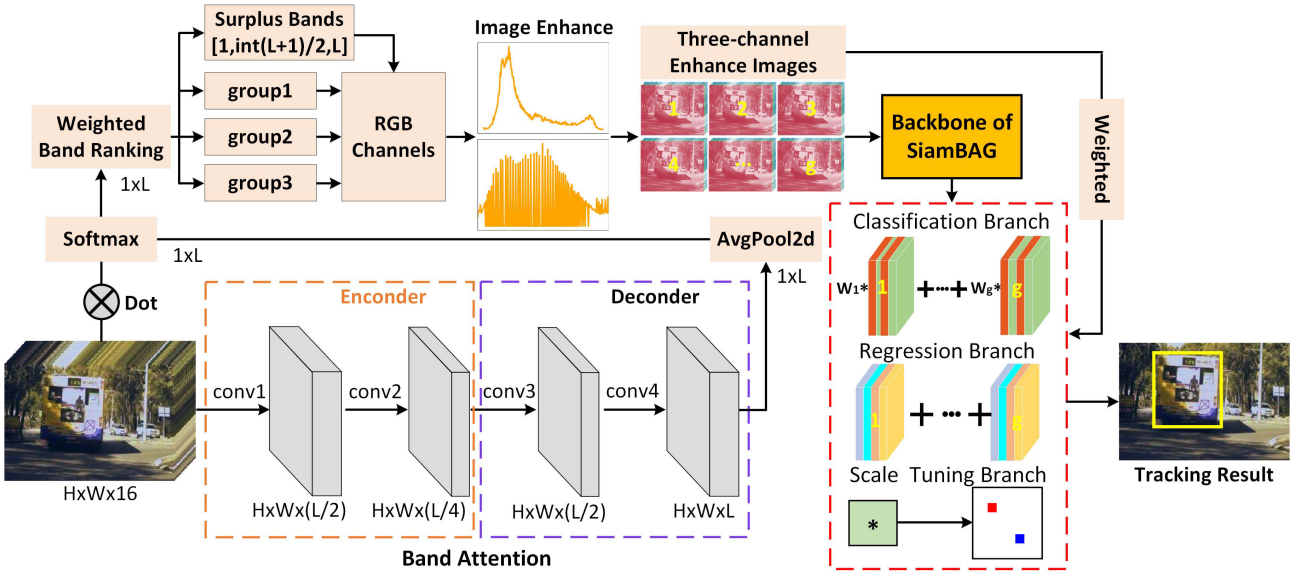


Fig. 1. Framework of the proposed SiamBAG method for hyperspectral object tracking. Key modules include the band attention network, band grouping strategy, image enhancement strategy, and the backbone network of SiamBAG.

information. Then they are processed by histogram equalization, finally forming multiple sets of enhanced three-channel images.

3) Instead of using ensemble learning that combines weak trackers for object localization, the classification responses corresponding to multiple classification branches are weighted for fusion, and then the fused results are used to perform scale regression calculation.
4) In the proposed SiamBAG network, a scale-tuning branch is added, which uses the similarity of the template frame and the detection frame to correct the raw position offsets, finally obtaining the fine-tuned object offsets.

The remainder of this article is organized as follows. In Section II, a detailed description of the proposed framework is presented. In Section III, hyperspectral videos are used to verify the proposed method, and the experimental results and parameters are discussed. Conclusions are drawn in Section IV.

II. SiamBAG Object Tracking Framework

As mentioned before, the main challenge faced by deep-learning-based methods in hyperspectral object tracking is the severe shortage of training data. Although collecting a large number of hyperspectral videos is a challenging task, there are massive existing RGB tracking datasets. In this section, we describe the proposed SiamBAG framework to leverage the RGB data for training. Essentially, SiamBAG is an integrated network, where an HSI is split into multiple three-channel false-color images according to their bandwise weight. After that, the SiamBAG network extracts color features for object representation in multiple false-color images, where massive RGB tracking data can be considered to train the tracking model.

Fig. 1 illustrates a flowchart of the proposed SiamBAG, which consists of four key modules. First, a band attention network is designed to obtain weighted rearranged bands. These bands are regrouped according to the corresponding weights; meanwhile, an image enhancement strategy is adopted to obtain multiple groups of three-channel enhanced false-color images. Then the enhanced false-color images are sent to the designed SiamBAG network for background classification and bounding boxes' regression, respectively. Finally, the regression results of different groups are fused by weight, and the bounding boxes with the maximum response are obtained. Meanwhile, the scale fine-tuning strategy is used to locate the tracking object in the detection frame.

A. Band Attention Network-Based Band Grouping

To adopt a tracker trained with a large number of RGB videos, an HSI needs to be decomposed into multiple groups of three-channel images. Because of the difference in material reflection, each band contains different information. Hence, it is very important to effectively divide these bands into multiple groups.

The ranking-based band selection methods provide a favorable manner to differentiate each band by assigning a corresponding weight. In the attention network [20], an attention model can take advantage of limited resources to focus on salient components of a signal. Hence, we can use the band-attention mechanism to adaptively rearrange bands by their importance. For an HSI $\mathbf{X} \in R^{N\times L}$ with N pixels and L bands, a band weighted pixel $\boldsymbol{y}_n$ is denoted by

$$\boldsymbol{y}_n = \boldsymbol{x}_n \odot \boldsymbol{w} \tag{1}$$

where $\boldsymbol{x}_n$ represents the original spectral vector, $\odot$ represents the elementwise multiplication, and $\boldsymbol{w} \in R^{1\times L}$ represents the bandwise weight. In the attention mechanism, the weight $\boldsymbol{w}$ is represented by

$$\boldsymbol{w} = f_\Theta(\boldsymbol{x}_n) \tag{2}$$

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where f_Θ is a mapping function that maps an input $\boldsymbol{x}_n$ to attention map, and Θ is a set of learnable parameters in an attention network.

Different from the "squeeze-and-excitation" block adopted by Hu et al. [29], the proposed method adaptively models the channelwise interdependencies with low computational cost. To this end, an autoencoder-like architecture [20], [22] is used to build the band attention network to exploit channel dependencies. As shown in Fig. 1, the encoder–decoder and average pooling correspond to "excitation" and "squeeze," where all the convolution operators are of size 1×1. The conv1 in the encoder shares the same parameters as conv4 in the decoder but in the transpose form, likewise for conv2 and conv3. This autoencoder-like architecture has fewer parameters than the "squeeze-and-excitation" block and is more conducive to learning the nonlinear correlation between bands. Moreover, the encoder is a dimensionality reduction module that can effectively learn the bandwise dependencies while suppressing the side effects of noises. In addition, the "squeeze" operator is set after the "excitation" operator because the global average pooling may result in loss of discrimination between different materials. For the original spectral vector $\boldsymbol{x}_n$, the band attention generation is described as

$$\boldsymbol{w} = \text{softmax}\left(f_{\text{pool}}\left(\sigma\left(\sigma(\boldsymbol{x}_n\boldsymbol{A}_1)\boldsymbol{A}_2\boldsymbol{A}_2^T\right)\boldsymbol{A}_1^T\right)\right) \tag{3}$$

where σ is the ReLU activation function. $\boldsymbol{A}_1$ and $\boldsymbol{A}_2$ are the parameters of conv1 and conv2, respectively. f_{pool} is an average pooling layer to fuse the band attention in each pixel. Then the attention map is generated using a softmax layer to normalize the weight between 0 and 1. Finally, the weight of channelwise is represented by

$$\boldsymbol{W} = [\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_L] \tag{4}$$

where $\boldsymbol{w}_L$ is the rearranged band weight. These weights meet the condition $\boldsymbol{w}_1 > \boldsymbol{w}_2, \ldots, > \boldsymbol{w}_L$. Likewise, the rearranged band set $\boldsymbol{B}_{\text{all}}$ is described as

$$\boldsymbol{B}_{\text{all}} = [\boldsymbol{B}_{i_1}, \boldsymbol{B}_{i_2}, \ldots, \boldsymbol{B}_{i_L}] \tag{5}$$

where $\boldsymbol{B}_{i_L}$ represents a single band with the band index i_L.

Since $\boldsymbol{B}_{\text{all}}$ is arranged in descending order according to the importance of bandwise, the closer the band is to the front, the more important the information it contains. To ensure that each group of the three-channel images has the approximate information as much as possible, a band grouping strategy is designed to split $\boldsymbol{B}_{i_L}$ into multiple three-channel images, which is formulated as

$$\boldsymbol{G}_{\text{all}} = [\boldsymbol{G}_1, \boldsymbol{G}_2, \ldots, \boldsymbol{G}_g] \tag{6}$$

where $\boldsymbol{G}_{\text{all}}$ is a set of all the three-channel images, $\boldsymbol{G}_g$ is the gth three-channel image, and g is the total number of the three-channel images calculated as

$$g = \begin{cases} \lfloor L/3 \rfloor, & \text{if } \mod(L,3) = 0 \\ \lfloor L/3 \rfloor + 1, & \text{otherwise} \end{cases} \tag{7}$$

where $\lfloor \cdot \rfloor$ is the floor operation, and $\mod(\cdot)$ is the remainder operation.

In detail, when $\mod(L,3) = 0$, $\boldsymbol{G}_{\text{all}}$ is first divided into three groups, which is formulated a

$$\begin{cases} \text{group1} = [\boldsymbol{B}_{i_1}, \boldsymbol{B}_{i_2}, \ldots, \boldsymbol{B}_{i_g}] \\ \text{group2} = [\boldsymbol{B}_{i_{g+1}}, \boldsymbol{B}_{i_{g+2}}, \ldots, \boldsymbol{B}_{i_{2*g}}] \\ \text{group3} = [\boldsymbol{B}_{i_{2*g+1}}, \boldsymbol{B}_{i_{2*g+2}}, \ldots, \boldsymbol{B}_{i_L}]. \end{cases} \tag{8}$$

Afterward, $\boldsymbol{B}_{\text{all}}$ is split into three subsets. Note that in group1, group2, and group3, the band weight is descended. Hence, all the weights in group1 are greater than those in group2; likewise, all the weights in group2 are greater than those in group3. To ensure that each group of the three-channel image has the approximate information as much as possible, the band index of $\boldsymbol{G}_i$ is obtained as

$$\boldsymbol{G}_i = [\text{group1}_i, \text{group2}_{g-i+1}, \text{group3}_i], \quad i = 1, 2, \ldots, g. \tag{9}$$

When $\mod(L,3)$ is not 0, the additional bands are merged into a new group. Hence, $\boldsymbol{G}_g$ is expressed as

$$\boldsymbol{G}_g = \begin{cases} [\boldsymbol{B}_{i_1}, \boldsymbol{B}_{i_{\lfloor L/2 \rfloor+1}}, \boldsymbol{B}_{i_L}], & \text{if } \mod(L,3) = 1 \\ [\boldsymbol{B}_{i_1}, \boldsymbol{B}_{i_{L-1}}, \boldsymbol{B}_{i_L}], & \text{if } \mod(L,3) = 2. \end{cases} \tag{10}$$

Afterward, an HSI is divided into multiple three-channel images.

To be specific, considering an HSI $\boldsymbol{X}$ containing 16 bands, if the band order is {15, 4, 6, 1, 11, 2, 3, 8, 7, 16, 13, 10, 5, 14, 9, 12}, group1 = {15, 4, 6, 1, 11}, group2 = {2, 3, 8, 7, 16}, and group3 = {13, 10, 5, 14, 9}. Hence, the resulted images are {X(:, [13], [15], [16]), X(:, [4], [7], [10]), X(:, [5], [6], [8]), X(:, [1], [3], [14]), X(:, [2], [9], [11]), and X(:, [7], [12], [15])}. Multiple images with approximate information are generated by this band attention grouping strategy. Then, they are processed by a histogram equalization [30] operation to form multiple enhanced three-channel false-color images and sent to the backbone of the SiamBAG network for object tracking.

B. Backbone of SiamBAG Network

As shown in Fig. 2, SiamBAG consists of a classification branch, a regression branch, and a scale-tuning branch. It takes SiamRPN [31] as the backbone network where a full convolution network without padding [31], [32] is used for feature extraction. It uses the modified AlexNet [33] as the backbone where groups from conv2 and conv4 are removed.

1) Backbone of SiamBAG: In the SiamBAG, the feature extraction consists of two branches, which are called the template branch and the detection branch, respectively. The template branch takes the tracking object patch in the previous frame as the input (denoted as $\boldsymbol{z}$), and the detection branch takes a tracking object patch in the current frame as the input (denoted as $\boldsymbol{x}$). Both the branches use the same AlexNet network for feature extraction, so the two patches are implicitly encoded by the same transformation. For convenience, $\phi(\boldsymbol{z})$ and $\phi(\boldsymbol{x})$ are used to represent the output feature maps of the Siamese subnetwork. Then, the features $\phi(\boldsymbol{z})$ and $\phi(\boldsymbol{x})$ are fed into the classification branch and regression branch

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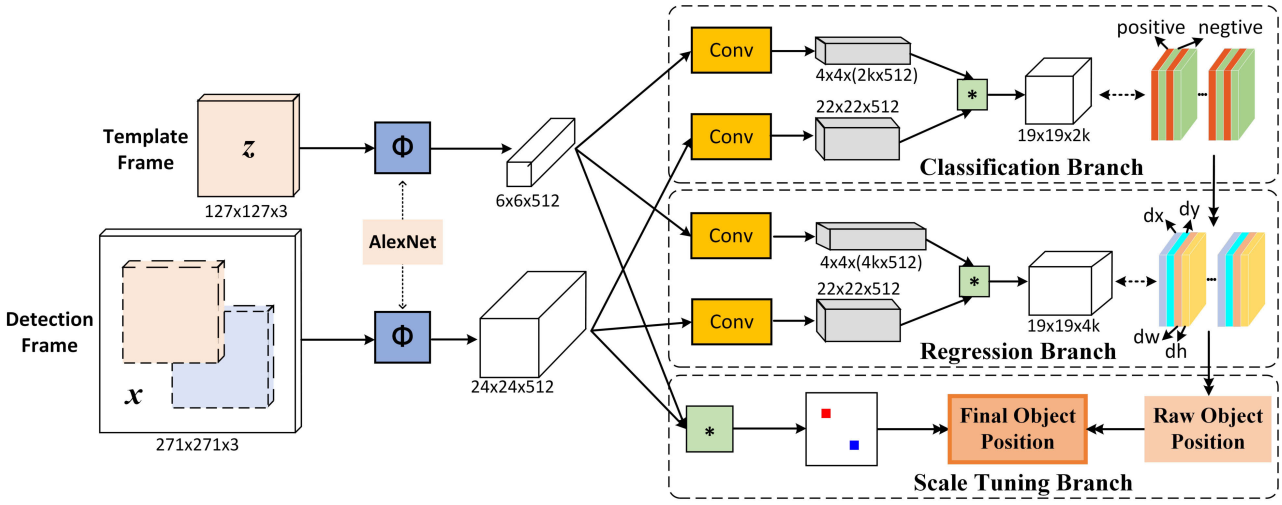


Fig. 2. Backbone of the proposed SiamBAG network includes a classification branch, a regression branch, and a scale-tuning branch.

simultaneously to obtain the classification response map and the regression response map, which are formulated as

$$\begin{cases} f^{\text{cls}}(\boldsymbol{x}, z) = [\phi(\boldsymbol{x})]_{\text{cls}} \star [\phi(z)]_{\text{cls}} \\ f^{\text{reg}}(\boldsymbol{x}, z) = [\phi(\boldsymbol{x})]_{\text{reg}} \star [\phi(z)]_{\text{reg}} \end{cases} \tag{11}$$

where $f^{\text{cls}}(\boldsymbol{x}, z)$ is the classification response map, $f^{\text{reg}}(\boldsymbol{x}, z)$ is the regression response map, and $\star$ is the convolution operation. The template feature maps $[\phi(z)]_{\text{cls}}$ and $[\phi(z)]_{\text{reg}}$ are used as kernels to perform the convolution operation.

In the proposed framework, the SiamRPN network is used as the baseline, and it is the off-the-shelf model provided by the corresponding authors [31]. Then, multiple enhanced three-channel images are sent to the backbone network to obtain multiple classification response maps $f^{\text{cls}}_{\mathbf{G}_i}(\boldsymbol{x}, z)$ and regression response maps $f^{\text{reg}}_{\mathbf{G}_i}(\boldsymbol{x}, z)$

$$\begin{cases} \left[f^{\text{cls}}_{\mathbf{G}_1}(\boldsymbol{x}, z), f^{\text{cls}}_{\mathbf{G}_2}(\boldsymbol{x}, z), \ldots, f^{\text{cls}}_{\mathbf{G}_g}(\boldsymbol{x}, z)\right] \\ \left[f^{\text{reg}}_{\mathbf{G}_1}(\boldsymbol{x}, z), f^{\text{reg}}_{\mathbf{G}_2}(\boldsymbol{x}, z), \ldots, f^{\text{reg}}_{\mathbf{G}_g}(\boldsymbol{x}, z)\right]. \end{cases} \tag{12}$$

As shown in Fig. 2, each point in $f^{\text{cls}}_{\mathbf{G}_i}$ with the size of $w \times h \times 2k$ is denoted as $(\bar{w}, \bar{h}, :)$, which contains a $2k$ channel vector, which represents for negative and positive activation of each anchor at the corresponding location on the original map. Softmax loss is adopted to supervise the classification branch. Similarly, each point in $f^{\text{reg}}_{\mathbf{G}_i}$ with the size of $w \times h \times 4k$ denoted as $(\tilde{w}, \tilde{h}, :)$ contains a $4k$ channel vector, which represents for dx, dy, dw, and dh measuring the distance between the anchor and the corresponding ground truth [31].

2) Fusion of Multiple Response Maps: After obtaining multiple sets of classification response maps, these classification response maps are fused with their weights, which are formulated as

$$\begin{cases} f^{\text{cls}}_{\mathbf{G}_{\text{all}}}(\boldsymbol{x}, z) = W_1 \cdot f^{\text{cls}}_{\mathbf{G}_1}(\boldsymbol{x}, z) + W_2 \cdot f^{\text{cls}}_{\mathbf{G}_2}(\boldsymbol{x}z) \\ \qquad\qquad + \cdots + W_g \cdot f^{\text{cls}}_{\mathbf{G}_g}(\boldsymbol{x}, z) \end{cases} \tag{13}$$

where $f^{\text{cls}}_{\mathbf{G}_{\text{all}}}(\boldsymbol{x}, z)$ is the final weighted fusion response map used to score the foreground–background classification. W_1, W_2, to W_g are the regularized weights of the false-color image, which are obtained by adding the weights of the respective three color channels.

In contrast, the offsets (dx, dy, dw, dh) calculated by multiple sets of regression response maps are considered to be equally important. Hence, the final regression response map $f^{\text{reg}}_{\mathbf{G}_{\text{all}}}(\boldsymbol{x}, z)$ is obtained by

$$f^{\text{reg}}_{\mathbf{G}_{\text{all}}}(\boldsymbol{x}, z) = \frac{1}{g} \sum_{i=1}^{g} f^{\text{reg}}_{\mathbf{G}_i}(\boldsymbol{x}, z). \tag{14}$$

3) Scale-Tuning Branch: As shown in Fig. 2, in the classification branch and regression branch, k is the number of anchors, where a region proposal network (RPN) [31] is used to determine the object size. dx and dy are the offsets of the center coordinates; likewise, dw and dh are the offsets of width and height. dx, dy, dw, and dh are used to measure the distances between the anchors and the corresponding ground truth.

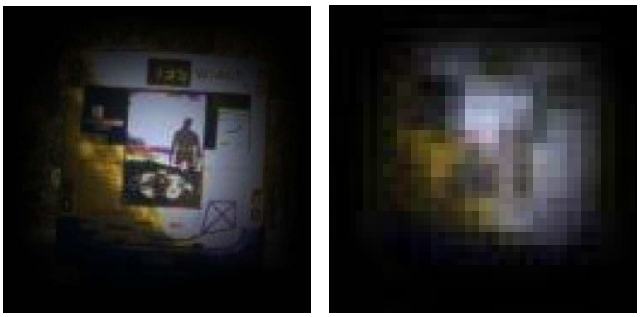
However, in Fig. 3, the size of the regression feature response map is 19×19, and the size of the detection frame map is 271×271. The scale ratio of the two maps is approximately 14 times, which leads to a deviation of approximately 14 pixels in the detection frame map shown in Fig. 3(a) when using the regression feature map shown in Fig. 3(b) for center positioning. The trend of scale variation is slowed down by the scale penalty strategy, but inevitably the center offsets (dx, dy) obtained by the feature response map still have some deviation, causing the center positioning error of the tracking object.

To reposition the center of the tracking object, the scale-tuning branch is proposed to correct the raw position offsets. In the scale-tuning branch, we use the similarity of the template frame and detection frame to fine-tune the offsets of the tracking object. It is formulated as

$$f^{\text{final}}(\boldsymbol{x}, z) = [\phi(\boldsymbol{x})]_{\text{raw}} \star [\phi(z)]_{\text{raw}} \tag{15}$$

where $f^{\text{final}}(\boldsymbol{x}, z)$ is the final response map used to locate the precise position of the target object. $[\phi(\boldsymbol{x})]_{\text{raw}}$ and $[\phi(z)]_{\text{raw}}$ are the output feature maps, which come from these patches

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(a) (b)

Fig. 3. Comparison of the detection frame map and regression feature response map. (a) Detection frame map with the size of 271 × 271. (b) Regression feature response map with the size of 19 × 19.

cut from the raw position. After obtaining the precisely located center coordinates, a scale pool is used to make the bounding box more compact with the relocated object, which is formulated as $\boldsymbol{S} = \{s_1, s_2, \ldots, s_m\}$, where s_m is the scale factor and m is the number of scale factors. The bounding box obtained by plenty of anchors is close to the ground truth, and hence, the scale pool is set to three sizes for fine-tuning, defined as $\boldsymbol{S} = \{0.98, 1, 1.02\}$. Note that these scale parameters can be adjusted according to the actual task. Finally, the maximum response is used to find its corresponding optimal scale factor to fine-tune the predicted bounding box.

4) Training and Loss Function: Band-attention-based image splitting embodies the importance of each group image. Different images play different roles in various scenes, which is also reflected by the band attention network due to its data-driven nature. Accordingly, the ensemble loss function of multiple three-channel images is defined as

$$L_{\text{bae}} = \frac{1}{g}\sum_{i=1}^{g} L_{\text{bae_}i} \tag{16}$$

where $L_{\text{bae_}i}$ is the same as that of BAENet [20], and all the parameters are trained in an end-to-end manner. More detailed descriptions can be found in [21] and [20].

Like the SiamRPN network, the loss function used in Faster R-CNN [34] is used to train the network with several anchors, where the cross-entropy loss and the smooth L_1 loss with normalized coordinates are used for classification and regression, respectively. Let A_x, A_y, A_w, and A_h denote the center point and shape of the anchor boxes, and let T_x, T_y, T_w, and T_h denote those of the ground-truth boxes likewise, and the normalized distance is calculated as

$$\begin{cases} \delta[0] = \dfrac{T_x - A_x}{A_w}, & \delta[1] = \dfrac{T_y - A_y}{A_h} \\ \delta[2] = \ln\dfrac{T_w}{A_w}, & \delta[3] = \ln\dfrac{T_h}{A_h}. \end{cases} \tag{17}$$

The smooth L_1 loss function is formulated as

$$\text{smooth}_{L_1}(x, \beta)\begin{cases} 0.5\beta^2 x^2, & |x| < \dfrac{1}{\beta^2} \\ |x| - \dfrac{1}{2\beta^2}, & |x| \geq \dfrac{1}{\beta^2} \end{cases} \tag{18}$$

where β is a threshold specified to change between L_1 and L_2 losses, which must be nonnegative. After they all pass through smooth L_1 loss, the final loss function is optimized by

$$\text{loss} = L_{\text{cls}} + \lambda L_{\text{reg}} \tag{19}$$

where L_{cls} is the cross-entropy loss, and λ is a hyperparameter to balance the two parts. L_{reg} is the loss of regression, which is calculated as

$$L_{\text{reg}} = \sum_{i=0}^{3} \text{smooth}_{L_1}(\delta[i], \beta) \tag{20}$$

where 3 represents the maximum subscript of normalized distance $\delta[i]$ in (17), with values ranging from 0 to 3. Since SiamBAG takes SiamRPN as the backbone network, the training process of the model is the same as SiamRPN. More detailed descriptions can be found in [31].

C. Compare With Related Work

In this article, the proposed SiamBAG tracker differs from the state-of-the-art BAENet [20] tracker. There are some essential differences. BAENet uses the hyperspectral dataset to train the model, while SiamBAG uses the color datasets to train the model. BAENet adopts online training and online testing for object tracking. Hence, it has extremely low tracking efficiency, and once the tracking object is lost, it cannot be retrieved. However, the SiamBAG uses offline training and online testing for object tracking, the tracking efficiency is high, and the object can be retrieved after being lost. BAENet rearranges these bands according to their weights, dividing them directly into multigroups, and discarding the last band, while SiamBAG divides these bands into multigroups without discarding the last band. First, all the bands are rearranged according to their weights from highest to lowest, and then they are divided into three major groups. Finally, the proposed band grouping strategy is performed to ensure that each group of three-channel images has the approximate total weight as much as possible. In addition, the last band is not deleted, and more band information is used as much as possible through supplementary means. BAENet uses the VITAL [21] tracker to track the object. SiamBAG first uses a histogram equalization strategy to enhance the features of multiple three-channel images and then uses the proposed backbone network to extract the object features. Meanwhile, the proposed scale fine-tuning branch is also used to relocate the center of the object. BAENet fuses multiple tracking results by directly summing them up. In SiamBAG, for different multiple three-channel images, first, the weights in the band attention network are used to fuse these multiple three-channel features at the feature level, and then the fused feature is used to locate the object.

In summary, our proposed SiamBAG is completely different from BAENet in terms of model training, band grouping strategy, feature extraction network, and result fusion strategy. Hence, it cannot be regarded as a naïve extension of BAENet by replacing the ensemble learning on object localization with a response map.

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III. Experimental Results and Discussion

To verify the performance of the proposed SiamBAG tracker, several trackers are taken as comparison methods. These include fast discriminative scale space tracker (fDSST) [35], background-aware correlation filter (BACF) [36], SiamFC [32], SiamRPN [31], BAENet [20], MHT [2], and TSCFW [16]. Extensive experiments are conducted on a benchmark hyperspectral/false-color video dataset [2]. In our experiments, all the parameters of the proposed model were set to the same as those in SiamRPN [31]. The parameters of the compared trackers followed settings in the corresponding papers.

All the experiments were performed on a desktop computer with a Windows 10 64-bit operating system, the Intel Core i7-8700H CPU with 3.20 GHz, 28 GB of main memory, and an NVIDIA GeForce GTX 1660Ti GPU.

A. Experimental Settings

In this article, the baseline hyperspectral video dataset is used to validate the performance of the proposed SiamBAG tracker, which is provided by Xiong et al. [2] in the hyperspectral object tracking challenge contest. These videos are captured at 25 frames/s (FPS) by an XIMEA snapshot camera, where each frame is originally captured in 2-D with 16 bands arranged in a mosaic mode, and the wavelength range is from 470 to 620 nm. These shared videos have been preprocessed, including image registration and spectral calibration. In the baseline hyperspectral video dataset, all the 35 test video scenes are used to verify the proposed method. Moreover, these video scenes cover 11 challenging factors, including background clutters (BCs), deformation (DEF), fast motion (FM), in-plane rotation (IPR), illumination variation (IV), low resolution (LR), motion blur (MB), occlusion (OCC), out-of-plane rotation (OPR), out-of-view (OV), and scale variation (SV). These basic attributes are listed in Table I for each video. More detailed descriptions can be found in [2] and [16].

In the proposed SiamBAG model, a modified AlexNet is used for features' extraction with the parameters of the first three convolution layers fixed and only fine-tuning the last two convolution layers. These parameters of the model are trained by optimizing the loss function in (19) with stochastic gradient descent (SGD).

Our proposed SiamBAG is trained on four extra color datasets without the HSIs. The first color dataset is the ImageNet Large Scale Visual Recognition Challenge for the VIdeo object Detection (ILSVRC-VID) dataset,[1] which is an image dataset organized according to the WordNet hierarchy. In ImageNet, images of each concept are quality-controlled and human-annotated. The second color dataset is the YouTube-Bounding Boxes (YouTube-BB) dataset,[2] which is a large-scale dataset of video URLs with densely sampled high-quality single-object bounding box annotations. Its goal with the public release of this dataset is to help advance the state of the art of machine learning for video understanding. The third color dataset is the ImageNet dataset.[3] Since 2010, the dataset is used in the ImageNet Large Scale Visual Recognition Challenge (ILSVRC), a benchmark in image classification and object detection. The publicly released dataset contains a set of manually annotated training images. A set of test images are also released, with the manual annotations withheld. The fourth color dataset is The Common Objects in Context (COCO) dataset,[4] which is widely used in computer vision research and has been used to train and evaluate many state-of-the-art object detection and segmentation models.

Furthermore, the training process of SiamBAG is the same as that of SiamRPN, where color images are involved in the training processing. Our proposed SiamBAG is implemented in Python under the framework of PyTorch. Note that these pretrained models[5,6] trained by original authors can be directly downloaded and migrated to SiamBAG for the hyperspectral object tracking task.

TABLE I
Basic Properties of Hyperspectral Video Scenes

Scenes	Frames	Size	Attributes
ball	625	471×207×16	SV,MB,OCC
basketball	186	463×256×16	FM,MB,OCC,LR
board	471	277×140×16	IPR,OPR,BC,OCC,SV
book	601	317×148×16	IPR,OPR,DEF
bus	131	261×148×16	LR,BC,FM
bus2	326	361×167×16	IV,SV,OCC,FM
campus	976	384×157×16	IV,SV,OCC
car	101	261×148×16	SV,OCC,IPR,OPR
car2	131	351×167×16	SV,IPR,OPR
car3	331	512×256×16	SV,LR,OCC,IV
card	930	363×200×16	IPR,BC,OCC
coin	149	291×120×16	BC
coke	731	493×207×16	BC,IPR,OPR,FM,SV
drive	725	297×142×16	BC,IPR,OPR,SV
excavator	501	500×240×16	IPR,OPR,SV,OCC,DEF
face	279	446×224×16	IPR,OPR,SV,MB
face2	1111	446×224×16	IPR,OPR,SV,OCC
forest	530	512×256×16	BC,OCC
forest2	363	512×256×16	BC,OCC
fruit	552	493×232×16	BC,OCC
hand	184	314×186×16	BC,SV,DEF,OPR
kangaroo	117	385×206×16	BC,SV,DEF,OPR,MB
paper	278	446×224×16	IPR,BC
pedestrain	306	351×167×16	IV,SV
pedestrain2	363	512×256×16	OCC,LR,DEF,IV
player	901	463×256×16	IPR,OPR,DEF,SV
playground	800	463×256×16	SV,OCC
rider1	336	512×256×16	LR,OCC,IV,SV
rider2	210	512×256×16	LR,OCC,IV,SV
rubik	526	493×207×16	DEF,IPR,OPR
student	396	438×256×16	IV,SV
toy1	376	271×135×16	BC,OCC
toy2	601	371×171×16	BC,OCC,SV,IV,OPR
truck	221	512×256×16	OCC,IV,SV,OV
worker	1209	228×121×16	SV,LR,BC

B. Evaluation Metrics

To evaluate the performance of various trackers, precision plot, success plot, and their corresponding area under curve (AUC) scores are used. Moreover, the precision rate (PR)

[1] https://www.image-net.org/challenges/LSVRC/
[2] https://research.google.com/youtube-bb/
[3] https://paperswithcode.com/dataset/imagenet
[4] https://www.v7labs.com/blog/coco-dataset-guide
[5] https://www.hsitracking.com
[6] https://github.com/huanglianghua/siamrpn-pytorch

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at the threshold of 20 pixels (DP@20) value is also used for performance assessment [2], [13]. All the results were recorded with one-pass evaluation (OPE).

In the precision plot, the horizontal axis represents the center location error (CLE), and the vertical axis represents the PR. The CLE denotes the center difference between the predicted bounding box and the ground truth, which is defined as

$$\mathrm{CLE} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2} \tag{21}$$

where (x_1, y_1) is the center position of the predicted bounding box, and (x_2, y_2) is the center location of the actual ground truth. The PR is defined as

$$\mathrm{PR}(\zeta) = \frac{N_{\mathrm{CLE}<\zeta}}{N_{\mathrm{total}}} \tag{22}$$

where ζ is a given threshold and its value is an integer from 0 to 50 pixels. When CLE $< \zeta$, a frame is considered a successfully tracked frame. N_{total} and $N_{\mathrm{CLE}<\zeta}$ represent the number of total frames and the number of frames with CLE less than ζ, respectively.

In the success plot, the horizontal axis represents the overlap threshold, and the vertical axis represents the success rate (SR). The intersection over union (IOU) between the predicted bounding box and the actual ground truth is defined as

$$\mathrm{IOU} = \frac{A_t \cap A_p}{A_t \cup A_p} \tag{23}$$

where A_p and A_t represent the area of the predicted bounding box and the ground truth, respectively. $\cap$ and $\cup$ represent the intersection and the union operations of two rectangular boxes, respectively. The SR is used to measure the IOU between the predicted bounding box and the actual ground truth, which is calculated as

$$\mathrm{SR}(\tau) = \frac{N_{\mathrm{IOU}>\tau}}{N_{\mathrm{total}}} \tag{24}$$

where τ is a threshold with a value from 0 to 1. When IOU is larger than τ, a frame is considered as a successfully tracked frame.

Taking CLE and PR(ζ) as the horizontal axis and vertical axis, respectively, the precision plot is obtained, where the closer the curve is to the upper left corner, the better the tracker performs. Similarly, taking overlap threshold τ and SR(τ) as the horizontal axis and vertical axis, respectively, the success plot is obtained, where the closer the curve is to the upper right corner, the better the tracker performs. In addition, their corresponding AUC values are also marked on the legend to rank all the trackers.

C. Results' Comparison

Some CF-based object tracking methods were further used for competitive comparison. Notably, BAENet [20], MHT [2], TSCFW [16], and SiamBAG were tested on hyerspectral videos, while fDSST [35], BACF [36], SiamFC [32], and SiamRPN [31] were run on the corresponding false-color videos.

Taking *basketball*, *coke*, *pedestrain2*, and *playround* scenes as examples, the visual tracking results of all the trackers are displayed in Fig. 4, where SiamBAG outperforms all others in the tracking performance, especially when it encounters FM (i.e., *basketball*), OCC (i.e., *pedestrain2*, *playround*), and severe nonrigid deformation (i.e., *coke*).

In Fig. 5, the experiment is conducted on 35 hyperspectral tracking videos, where the precision plots and success plots are used to evaluate the performance of various trackers. In Fig. 5(a), the black precision plot representing SiamBAG is the closest to the upper left corner, which indicates that it achieves the best tracking performance among these trackers. Similarly, in Fig. 5(b), the black success plot representing SiamBAG is also the closest to the upper right corner, and hence, the SiamBAG tracker achieves the best success tracking accuracy.

In Fig. 5, these precision plots and success plots of all the competing trackers on all the video scenes are also quantified by their corresponding AUC values. In these CF-based trackers, MHT and TSCFW achieve better tracking performance than fDSST and BACF. The reason is that the spectral feature of materials is considered in MHT and TSCFW trackers. In the MHT tracker, the SSHMG feature extractor and abundance information of material are used to obtain the object's spatial–spectral features. In the TSCFW tracker, the context-aware information and tensor representation of the tracking object are considered to obtain discriminative features. Therefore, when facing challenging scenarios, the tracking performance of MHT and TSCFW is effectively stimulated, resulting in more stable tracking effects. In these deep-learning-based trackers, the SiamFC gives inferior precision, due to the fact that it only learns a template by insufficient training data. When encountering complex changing scenarios, the convolution feature of the tracking object is disturbed, resulting in tracking failure. Due to the intrinsic limitations, BAENet needs to train and update the template online, making it feasible to use less training data for tracking tasks. This also inevitably results in large computational consumption. In contrast, SiamBAG and SiamRPN trackers obtain more negative samples and robust object features for template training using massive existing color object tracking data. Therefore, better tracking performance is obtained.

In Table II, the AUC values of success plots, DP@20, and frames per second (FPS) are listed. By comparing these quantitative results, it can be confirmed that the proposed SiamBAG produces the most appealing performance. Compared with BAENet, the proposed SiamBAG has a 3.7% improvement in the AUC of success plot and a 3% improvement in DP@20. However, the running cost of SiamBAG is mediocre. The reason is that the feature extraction of hyperspectral videos with more channels is time-consuming. In addition, real-time band grouping and scale fine-tuning are also computationally expensive in the proposed SiamBAG model.

To better highlight the advantages of the proposed SiamBAG tracker, in Table III, the AUC values for all the comparison trackers in different challenges are listed, where the tracking performance under different hyperspectral video attributes is analyzed. In our experiment, all the hyperspectral videos containing the same challenging factor are used for attribute evaluation. It is observed that in the SV, MB,

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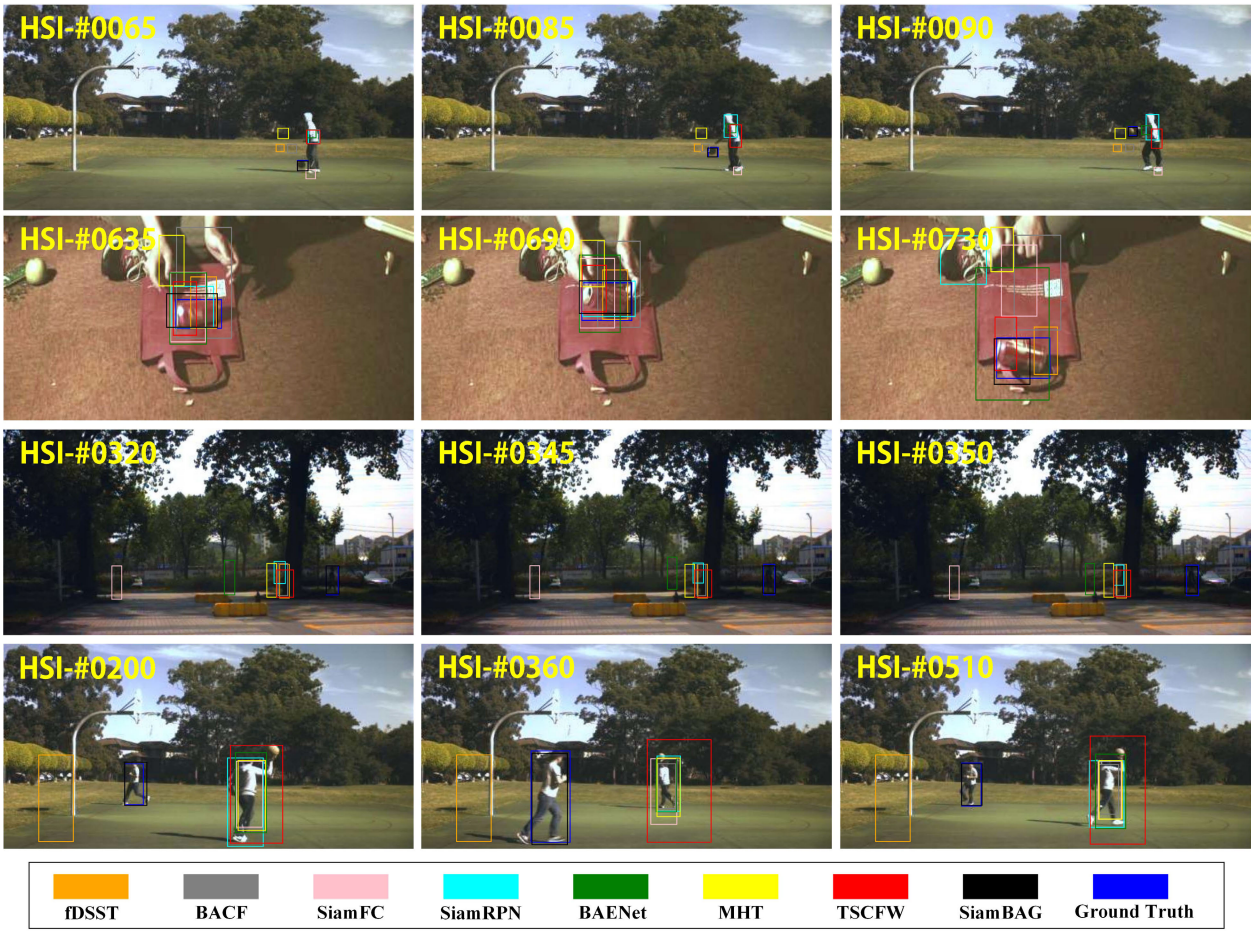
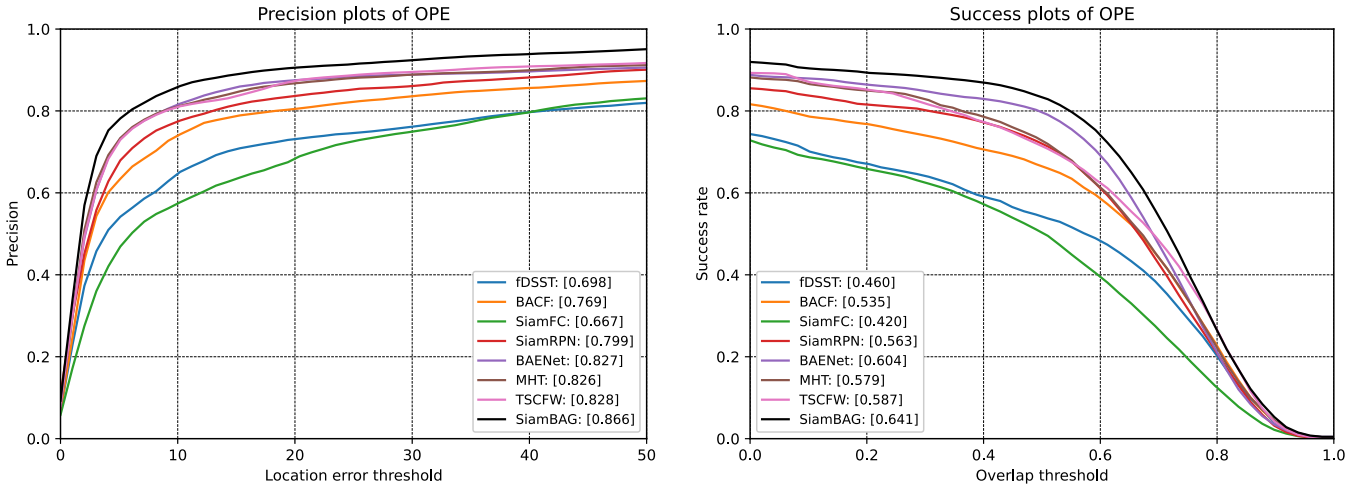


Fig. 4. Visualization of tracking results on four video sequences (i.e., *basketball, coke, pedestrain2, and playground*). Bounding boxes in different colors are the results of different trackers.



(a) (b)

Fig. 5. Comparison of various trackers. AUCs of these trackers are specified in the legends. (a) Precision plots of OPE. (b) Success plots of OPE.

OCC, LR, and DEF attributes, SiamBAG achieves the optimal performance, while the suboptimal performance is achieved under other attributes. These results further verify the superior performance of the proposed SiamBAG in hyperspectral video object tracking.

Furthermore, the tracking performance of each scene is also listed in Table IV, where mean IOU (mIOU) is added for quantitative comparison. mIOU is slightly different from AUC values of success plots, which is calculated by mean(IOU). Here, GOT-10k [37] evaluation protocol is used to calculate the

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TABLE II

OVERALL PERFORMANCE COMPARISON OF VARIOUS TRACKERS IN RGB AND HYPERSPECTRAL/FALSE-COLOR VIDEOS. THE BEST AND SECOND BEST VALUES ARE HIGHLIGHTED IN RED AND BLUE FONTS, RESPECTIVELY

Datasets	Methods	fDSST [35]	BACF [36]	SiamFC [32]	SiamRPN [31]	BAENet [20]	MHT [2]	TSCFW [16]	SiamBAG
RGB	AUC	0.486	**0.521**	0.416	**0.637**	n/a	n/a	n/a	n/a
	DP@20	0.707	**0.773**	0.667	**0.904**	n/a	n/a	n/a	n/a
	FPS	**135.863**	45.361	**98.513**	46.105	n/a	n/a	n/a	n/a
HSI/False-color	AUC	0.460	0.535	0.420	0.563	**0.604**	0.579	0.587	**0.641**
	DP@20	0.729	0.803	0.675	0.834	**0.874**	0.866	0.870	**0.904**
	FPS	**134.290**	44.827	**100.511**	46.344	0.697	2.253	3.620	5.740

TABLE III

ATTRIBUTE-BASED COMPARISON ON HYPERSPECTRAL/FALSE-COLOR VIDEOS. THE BEST AND SECOND BEST VALUES ARE HIGHLIGHTED IN RED AND BLUE FONTS, RESPECTIVELY. OUR TRACKER RANKS FIRST ON 5 OUT OF 11 ATTRIBUTES, I.E. SV, MB, OCC, LR, AND DEF, AND THE SECOND BEST IN OTHER ATTRIBUTES

Attributes	fDSST [35]*	BACF [36]*	SiamFC [32]*	SiamRPN [31]*	BAENet [20]	MHT [2]	TSCFW [16]	SiamBAG
SV	0.445	0.536	0.451	0.545	**0.597**	0.552	0.567	**0.636**
MB	0.574	**0.668**	0.504	0.654	0.623	0.605	0.613	**0.674**
OCC	0.453	0.526	0.446	0.541	0.564	**0.567**	0.545	**0.612**
FM	**0.611**	0.551	0.511	0.582	0.595	0.525	0.583	**0.602**
LR	0.337	0.340	0.297	0.389	0.523	0.493	**0.533**	**0.605**
IPR	0.620	0.674	0.464	0.670	0.705	0.680	**0.728**	**0.715**
OPR	0.578	**0.704**	0.502	0.665	**0.704**	0.641	0.692	**0.699**
DEF	0.602	0.672	0.448	**0.706**	0.686	0.682	0.672	**0.710**
BC	0.387	0.512	0.360	0.525	**0.637**	0.579	0.620	**0.635**
IV	0.337	0.468	0.373	0.446	**0.511**	0.416	0.454	**0.508**
OV	0.470	0.543	0.404	0.592	0.517	0.626	**0.654**	**0.635**

* denotes the corresponding false-color videos is used for evaluation.

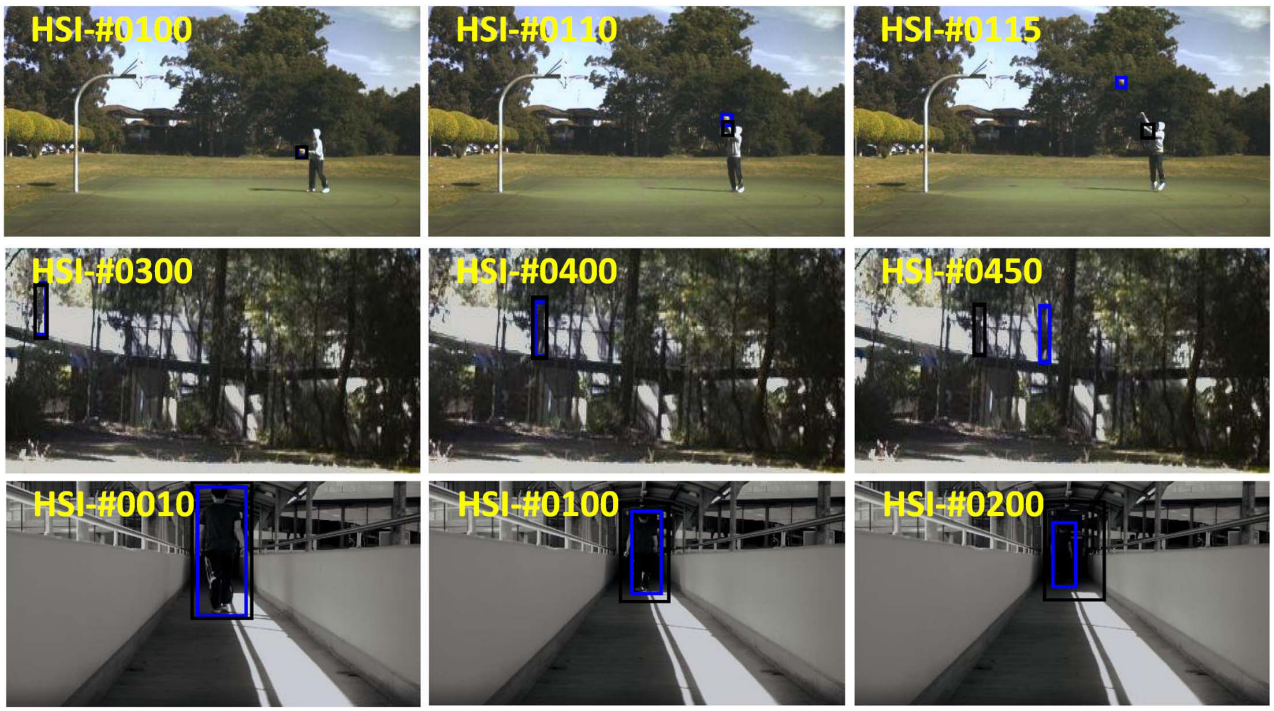


Fig. 6. Failure cases of our method (i.e., *basketball*, *campus*, and *student*). Blue and black bounding boxes denote the ground truths and our tracking results, respectively.

mIOU value. The GOT-10k toolkit is a visual tracking toolkit that implements evaluation metrics and tracking pipelines for different object tracking datasets.

As listed in Table IV, the AUC value of the proposed SiamBAG in six scenarios including *basketball*, *camp*, *rider1*, *student*, *toy1*, and *toy2* is lower than 0.5. In addition, Fig. 6 shows a few failure cases of the proposed SiamBAG tracker, where drastic illumination variation in target appearance change causes a drift problem in *campus* and *student* sequences, and fast movement also makes the proposed

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TABLE IV
SIAMBAG'S TRACKING PERFORMANCE UNDER DIFFERENT HYPERSPECTRAL VIDEO SCENES

Scenes	Frames	mIOU	AUC	DP@20	FPS
ball	625	0.697	0.687	0.987	5.7
basketball	186	0.436	0.431	0.608	5.7
board	471	0.665	0.656	0.813	5.8
book	601	0.800	0.790	1	5.8
bus	131	0.773	0.764	1	5.9
bus2	326	0.695	0.684	1	5.8
campus	976	0.448	0.441	0.677	5.8
car	101	0.758	0.749	1	5.8
car2	131	0.563	0.555	0.748	5.8
car3	331	0.558	0.550	0.903	5.7
card	930	0.661	0.651	1	5.8
coin	149	0.784	0.776	1	5.8
coke	731	0.589	0.579	0.925	5.7
drive	725	0.789	0.780	1	5.8
excavator	501	0.526	0.516	0.605	5.5
face	279	0.813	0.803	1	5.7
face2	1111	0.828	0.818	1	5.7
forest	530	0.735	0.725	1	5.7
forest2	363	0.623	0.616	0.796	5.7
fruit	552	0.570	0.562	0.837	5.7
hand	184	0.691	0.681	0.967	5.7
kangaroo	117	0.690	0.679	1	5.8
paper	278	0.776	0.766	1	5.7
pedestrain	306	0.604	0.595	1	5.8
pedestrain2	363	0.703	0.693	0.978	5.7
player	901	0.804	0.793	1	5.7
playground	800	0.727	0.716	0.999	5.7
rider1	336	0.467	0.460	0.714	5.7
rider2	210	0.531	0.525	0.662	5.7
rubik	526	0.700	0.690	1	5.7
student	396	0.292	0.281	1	5.6
toy1	376	0.360	0.354	0.524	5.8
toy2	601	0.480	0.473	0.621	5.7
truck	221	0.600	0.622	0.860	5.7
worker	1209	0.668	0.658	0.999	5.8
Total	**16574**	**65.3%**	**64.1%**	**90.4%**	**5.7**

TABLE V
ABLATION STUDY ON BAND ATTENTION GROUPING AND SCALE-TUNING BRANCH

Categories	AUC	ΔAUC	DP@20	ΔDP@20	FPS
both	0.641	+0.52	0.904	+0.60	5.7
with grouped	0.623	+0.34	0.877	+0.33	7.6
with scale tuning	0.605	+0.16	0.858	+0.14	6.6
neither	0.589	n/a	0.844	n/a	9.2

SiamBAG tracker miss the target completely in *basketball* sequence.

D. Ablation Analysis

The results of the ablation study on band attention grouping and scale-tuning branch are listed in Table V. The results demonstrate that the SiamBAG tracker has the most inferior performance when neither operation is carried out. When adding any of the two operations, the tracking performance of SiamBAG is improved. Obviously, when both the operations are implemented, the proposed SiamBAG produces the optimal tracking performance. Compared with neither operation, after both the operations are carried out, the proposed SiamBAG has 5.2% and 6.0% improvement in the AUC of success plot and DP@20, respectively. Both the band attention grouping and scale tune branch can improve the SiamBAG tracking performance. Moreover, from the results listed in Table V, the band attention grouping operation is more conducive to the improvement of accuracy than the scale fine-tuning strategy. The results further verify that two operations significantly improve the tracking performance in hyperspectral videos.

In terms of execution time, when neither operation is implemented, the FPS is the best, but its AUC value is reduced. When performing the band attention grouped operation, SiamBAG achieves better tracking efficiency than performing scale fine-tuning operation, and with better AUC value, which also meets the needs of practical application. The scale fine-tuning operation is more time-consuming, which is the main reason for the reduced tracking efficiency.

TABLE VI
ABLATION STUDY ON DIFFERENT IMAGE ENHANCEMENT STRATEGIES

Enhance Strategies	Global	Local
without enhance strategy	0.605/0.858	0.605/0.858
adaptive histogram equalization	0.610/0.869	**0.636/0.894**
global histogram equalization	**0.641/0.904**	0.611/0.876

Furthermore, the influence of different image enhancement strategies on tracking performance is also discussed. In Table VI, the results with and without image enhance strategy are listed. Obviously, the proposed SiamBAG with the image enhancement strategy outperforms the tracker without the image enhancement strategy in AUC and DP@20. Hence, these results verify that using the image enhancement strategy to process grouped three-channel images can significantly improve the robustness of the template. When the global histogram equalization operation is performed, the proposed SiamBAG achieves the best tracking performance.

IV. CONCLUSION

In this article, a novel band attention grouping-based Siamese framework has been proposed to handle the shortage of training data issues for the hyperspectral object tracking tasks. Using the existing massive color data to train the proposed SiamBAG network, the generalization and versatility of this model are effectively improved, which further confirmed that the SiamBAG network can effectively use color data to perform the hyperspectral object tracking task. In addition, the proposed attention-based band grouping and scale-tuning branch strategies have also been confirmed to effectively improve the information balance of multiple three-channel images and the accuracy of object localization. Experiments on the benchmark dataset have confirmed the performance of the proposed SiamBAG. However, the feature extraction from multiple three-channel images results in high computational costs. In the future, we will further reduce the computational cost of trackers for the requirement of real applications.

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